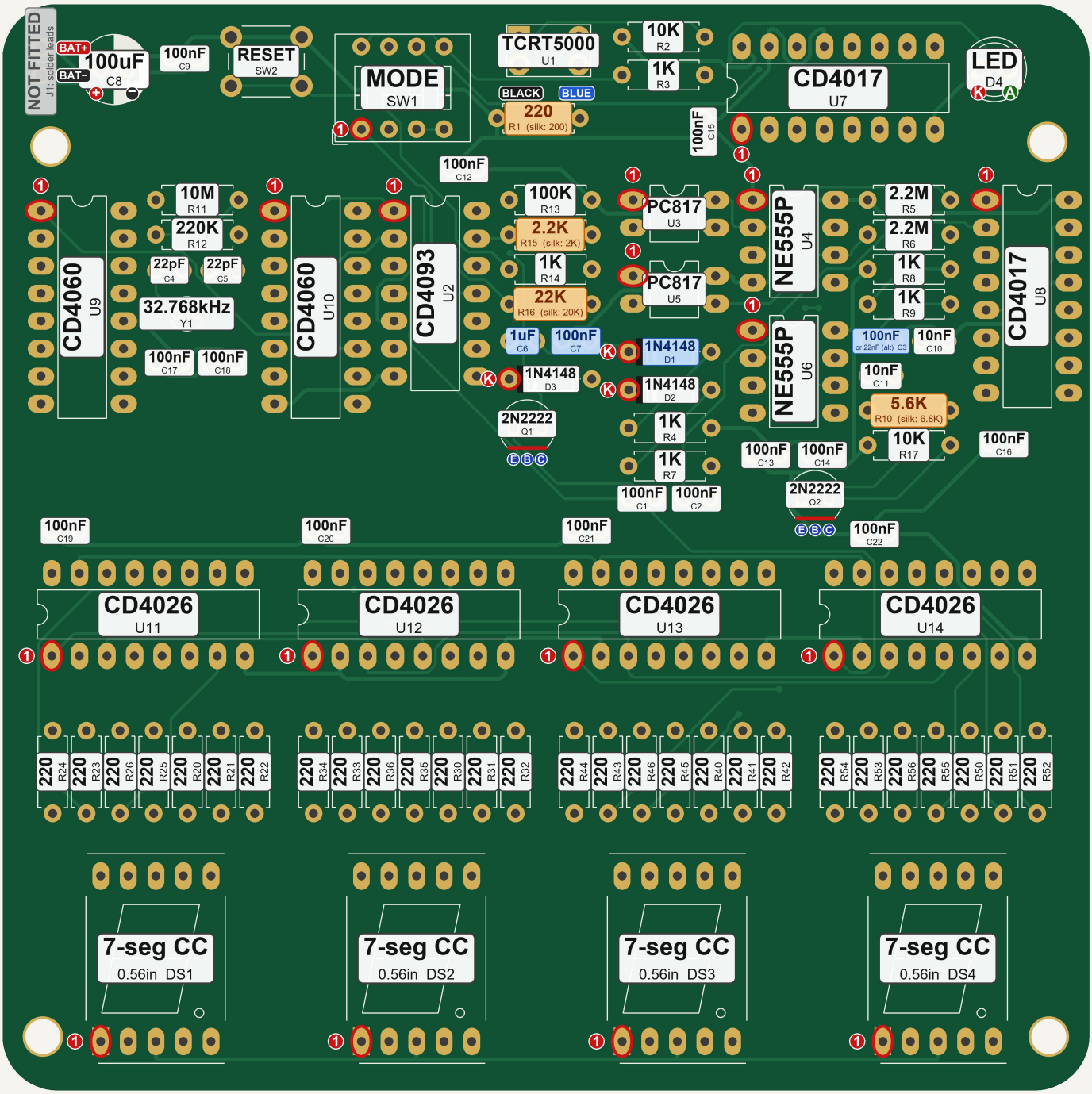


tach_v1.0 — fitted parts (from fab/tach_v1.0_bom_updated.csv)

- value as fitted (matches the silkscreen)
- SWAPPED from your stock — silkscreen shows the drawn value
- no value printed next to it on the silkscreen
- not fitted

Orientation / polarity

- Pin 1 (red ring). ICs: notch / dot end. PC817: dot on the package. SW1: switch "1". Displays: DP dot bottom-right.
- Diode cathode = black band (1N4148). Band goes on the ringed pad.
- LED D4 cathode: short leg, flat edge of the rim. A (green) = long leg.
- C8 electrolytic: long leg into "+"; the printed stripe (−) faces the other pad.
- 2N2222 (Q1, Q2): flat face on the red line, legs E-B-C left to right as shown.
- J1 (not fitted): battery + lead into the hole marked BAT+, − lead into BAT−.
- TCRT5000: black lens (sensor) over BLACK, blue/clear lens (IR LED) over BLUE.

No polarity: resistors, ceramic capacitors, crystal Y1, push button SW2.
Top view, component side. Each tag: fitted value (large), reference (small).